WEEK 1 ASSIGNMENT

Please follow the instructions below carefully and complete all tasks.

1. Server Access

Log into the server using SSH on port 22.
Host:

```
159.65.222.96
```

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2. User Creation

Create a Linux user account using the following format:
- Username: your first name + surname initial
- Example: If your name is Jane Doe → **JaneD**

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3. PostgreSQL Setup

Check if PostgreSQL is installed:

```
psql --version
```

If PostgreSQL is not installed, install it.

If it is installed, ensure the PostgreSQL service is running and accessible.

Also ensure that:

- The server allows external connections on port **5432**
- Users can successfully connect remotely to PostgreSQL

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4. Database Setup

- Create a database using the same name as your Linux username
- Create a schema called **staging**
- Generate sample data of your choice (any dataset you prefer)
- Upload/insert the data into the **staging schema** in your database

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5. Linux Commands Practice

Perform at least **25 Linux commands** and document them.
Include:
- Screenshots of each command and output

- Usage of **scp** to:
    - Transfer files from local PC to server
    - Transfer files from server to local PC

Other useful Linux commands (navigation, permissions, networking, process management, etc.)

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6. GitHub Documentation

- Create a well-organized GitHub repository or GitHub Pages site
- Publish your work professionally

Include:

- Steps performed
- Commands used
- Screenshots
- Explanations where necessary

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7. Blog Article

Write a technical article titled:

"Linux Fundamentals for Data Engineering"

Requirements:

- Minimum 1500 words
- Must be well-structured and professional
- Publish it on any blogging platform of your choice (Medium, Dev.to, Hashnode, etc.)
- Include practical examples from your assignment

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8. Submission

Send your blog link to:
dataengineering@luxdevhq.com

CC: mbaabuharun8@gmail.com

Making a secure connection to the server

On the terminal we will ssh the name of our server and the ip address.

```
C:\Users\USER>ssh root@159.65.222.96
root@159.65.222.96's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-71-generic x86_64)
```

Check the version of postgresql

It's important to check the version of the postgres.

```
  command 'psi' from deb psi (1.5+dfsg1-1.1)
Try: apt install <deb name>
root@AssignmentServer:~# psql  --version
psql (PostgreSQL) 16.14 (Ubuntu 16.14-0ubuntu0.24.04.1)
root@AssignmentServer:~# sudo systemctl status
```

checking the status of postgresql

The status is active.

```
root@AssignmentServer:~# history
   70  systemctl start postgresql
   71  systemctl enable postgresql
   72  systemctl status postgresql
```

Database set up

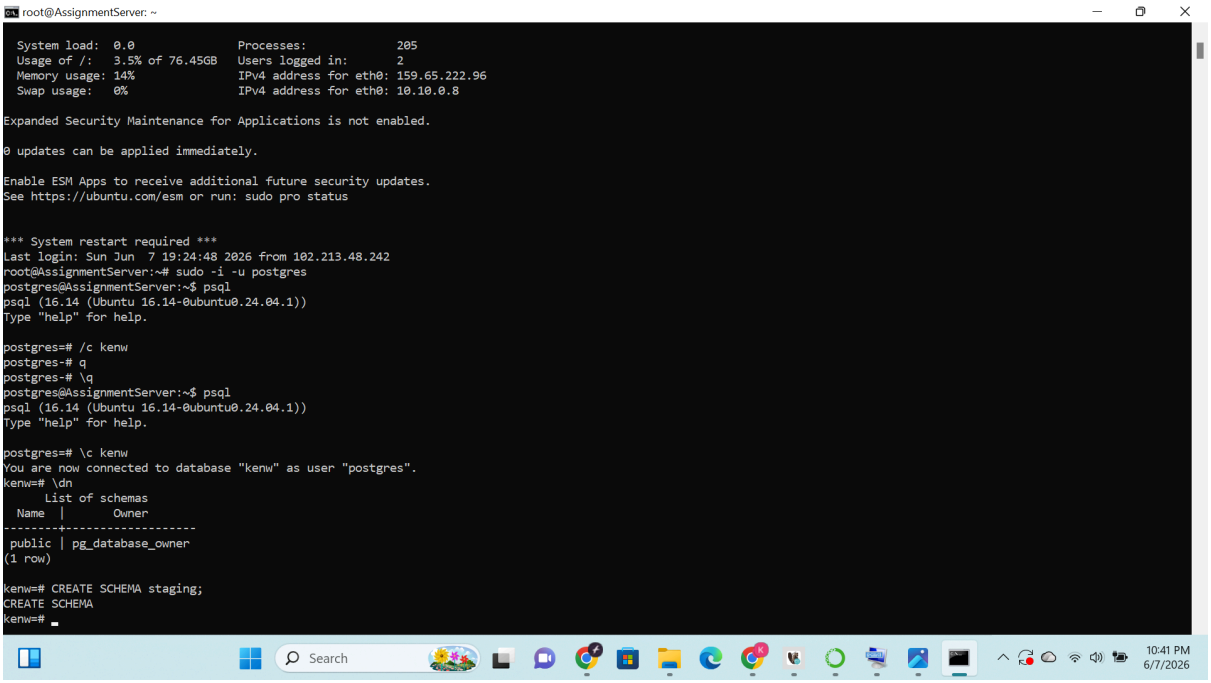
At the root user use this command *sudo -i -u postgres*

Later use *psql* to get into postgres and create a data base .

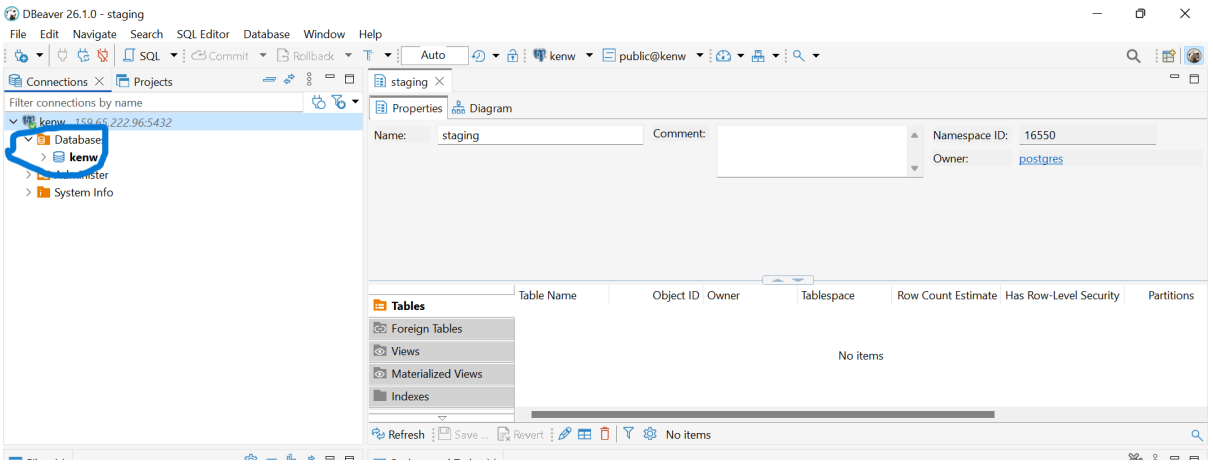
```
kenw@AssignmentServer:~$ cd ..
kenw@AssignmentServer:/home$ exit
exit
```

Create a schema called staging

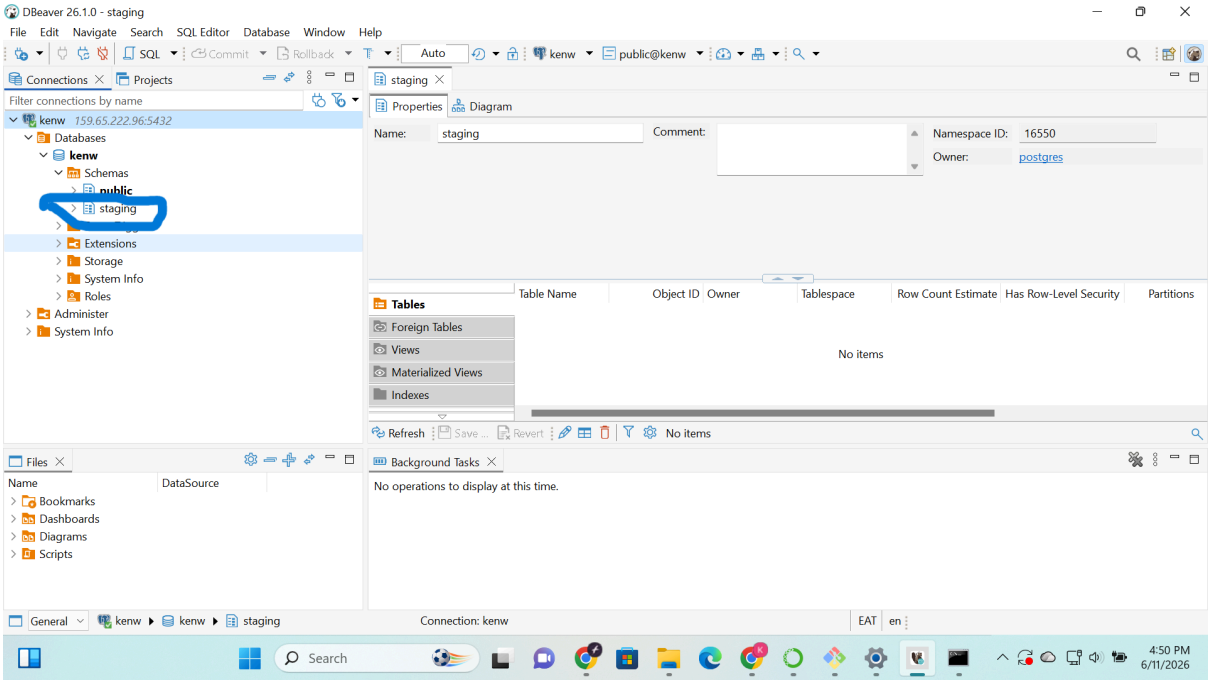
Creating a schema using the terminal.



Database on dbeaver



schema page



sending files from the computer to the server



Linux commands

scp (Secure Copy Protocol) command.
It's used to send files from the local machines to the server.

It's syntax

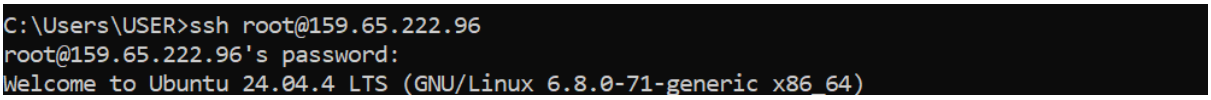


Adding a *hyphen* with an *r* ensures recursive sending of files in a folder.



SSH

ssh is known as Secure Shell, it is used to establish secure connection to a server.

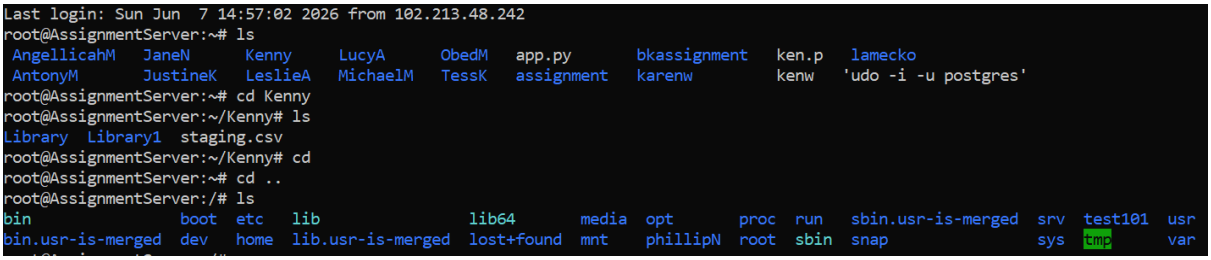


cd change directory

It's used to move from one directory to another.

ls list

It's used to list files in a directory.

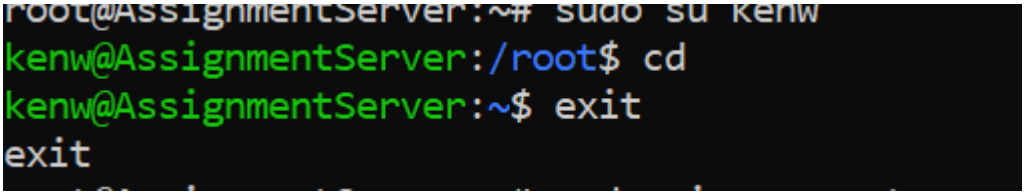


cd

It's used to move to the home of the user.

exit

It's used to exit from one user to the root user or exit from the root user to the normal terminal.



cd ..

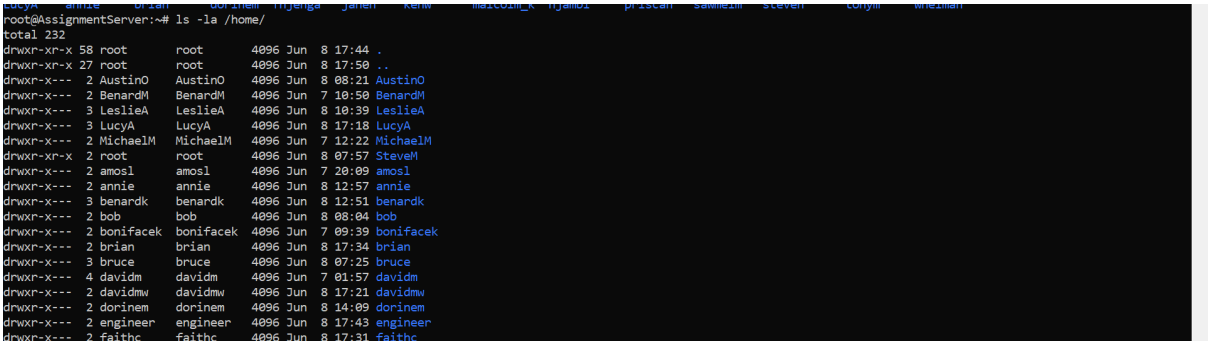
It's used up and down in a directory.



```
root@AssignmentServer:~# sudo su kenw
kenw@AssignmentServer:/root$ cd
kenw@AssignmentServer:~$ exit
exit
```

ls -lh

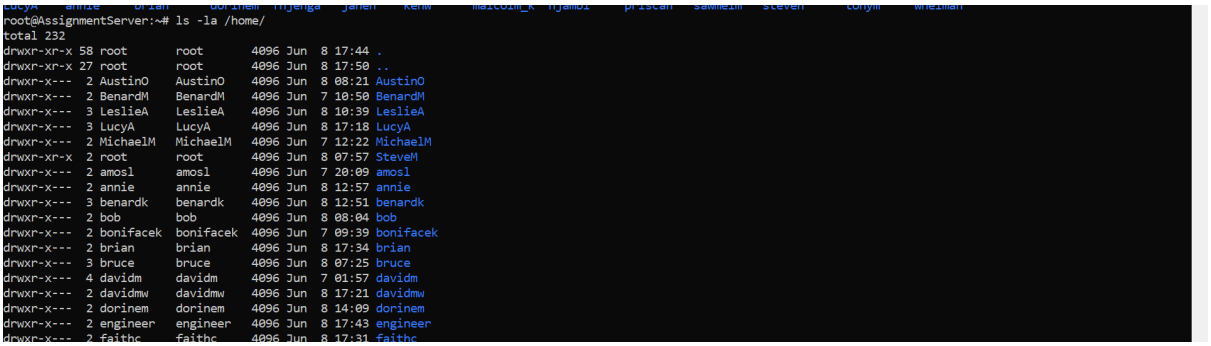
It's used to list all files or directory in a long format showing the type of file,its owner,user,the date it was created and its last modification date.



```
root@AssignmentServer:~# ls -la /home/
total 232
drwxr-xr-x 58 root       root       4096 Jun  8 17:44 .
drwxr-xr-x 27 root       root       4096 Jun  8 17:50 ..
drwxr-x---  2 AustinO    AustinO    4096 Jun  8 08:21 AustinO
drwxr-x---  2 BenardM    BenardM    4096 Jun  7 10:50 BenardM
drwxr-x---  3 LeslieA    LeslieA    4096 Jun  8 10:39 LeslieA
drwxr-x---  3 LucyA      LucyA      4096 Jun  8 17:18 LucyA
drwxr-x---  2 MichaelM   MichaelM   4096 Jun  7 12:22 MichaelM
drwxr-xr-x  2 root       root       4096 Jun  8 07:57 SteveM
drwxr-x---  2 amosl      amosl      4096 Jun  7 20:09 amosl
drwxr-x---  2 annie      annie      4096 Jun  8 12:57 annie
drwxr-x---  3 benardk    benardk    4096 Jun  8 12:51 benardk
drwxr-x---  2 bob        bob        4096 Jun  8 08:04 bob
drwxr-x---  2 bonifacek  bonifacek  4096 Jun  7 09:39 bonifacek
drwxr-x---  2 brian      brian      4096 Jun  8 17:34 brian
drwxr-x---  3 bruce      bruce      4096 Jun  8 07:25 bruce
drwxr-x---  4 davidm     davidm     4096 Jun  7 01:57 davidm
drwxr-x---  2 davidmw    davidmw    4096 Jun  8 17:21 davidmw
drwxr-x---  2 dorinem    dorinem    4096 Jun  8 14:09 dorinem
drwxr-x---  2 engineer   engineer   4096 Jun  8 17:43 engineer
drwxr-x---  2 faithc     faithc     4096 Jun  8 17:31 faithc
```

ls -la

It's used to show long format all the hidden files .



```
root@AssignmentServer:~# ls -la /home/
total 232
drwxr-xr-x 58 root       root       4096 Jun  8 17:44 .
drwxr-xr-x 27 root       root       4096 Jun  8 17:50 ..
drwxr-x---  2 AustinO    AustinO    4096 Jun  8 08:21 AustinO
drwxr-x---  2 BenardM    BenardM    4096 Jun  7 10:50 BenardM
drwxr-x---  3 LeslieA    LeslieA    4096 Jun  8 10:39 LeslieA
drwxr-x---  3 LucyA      LucyA      4096 Jun  8 17:18 LucyA
drwxr-x---  2 MichaelM   MichaelM   4096 Jun  7 12:22 MichaelM
drwxr-xr-x  2 root       root       4096 Jun  8 07:57 SteveM
drwxr-x---  2 amosl      amosl      4096 Jun  7 20:09 amosl
drwxr-x---  2 annie      annie      4096 Jun  8 12:57 annie
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drwxr-x---  2 bob        bob        4096 Jun  8 08:04 bob
drwxr-x---  2 bonifacek  bonifacek  4096 Jun  7 09:39 bonifacek
drwxr-x---  2 brian      brian      4096 Jun  8 17:34 brian
drwxr-x---  3 bruce      bruce      4096 Jun  8 07:25 bruce
drwxr-x---  4 davidm     davidm     4096 Jun  7 01:57 davidm
drwxr-x---  2 davidmw    davidmw    4096 Jun  8 17:21 davidmw
drwxr-x---  2 dorinem    dorinem    4096 Jun  8 14:09 dorinem
drwxr-x---  2 engineer   engineer   4096 Jun  8 17:43 engineer
drwxr-x---  2 faithc     faithc     4096 Jun  8 17:31 faithc
```

echo

It's used to create a file.

Strings in this example have been stored in *app.py*

cat

It's used to read file contents.

```
navas@AssignmentServer:~$ ls
navas@AssignmentServer:~$ mkdir project
navas@AssignmentServer:~$ ls
project
navas@AssignmentServer:~$ cd project/
navas@AssignmentServer:~/project$ ls
navas@AssignmentServer:~/project$ echo "Hello world!"
Hello world!
navas@AssignmentServer:~/project$ echo "Hello world!" > app.py
navas@AssignmentServer:~/project$ ls
app.py
navas@AssignmentServer:~/project$ cat app.py
Hello world!
navas@AssignmentServer:~/project$ cp app.py app_backup.py
navas@AssignmentServer:~/project$ ls
app.py  app_backup.py
navas@AssignmentServer:~/project$ cat app_backup.py
Hello world!
navas@AssignmentServer:~/project$ ls -l
total 8
-rw-rw-r-- 1 navas navas 13 Jun  9 07:39 app.py
-rw-rw-r-- 1 navas navas 13 Jun  9 07:40 app_backup.py
navas@AssignmentServer:~/project$ cd ..
```

history

It's used to show all the commands we have used in our work.

```
root@AssignmentServer:~# history
   70  systemctl start postgresql
   71  systemctl enable postgresql
   72  systemctl status postgresql
```

mkdir

It's used in creating a new directory/folder.

```
kenw@assigmentServer:~$ ls
KennedyW  amin.py  amout.py  kenny  thehub
kenw@assigmentServer:~$ mkdir amhim
kenw@assigmentServer:~$ ls
KennedyW  amhim  amin.py  amout.py  kenny  thehub
```

pwd

It's in the printing working directory,it helps in showing one the current directory one is working on.

```
kenw@assigmentServer:~$ pwd
/home/kenw
kenw@assigmentServer:~$
```

\q
It is used to quit without saving a file (:q!).

With a front slash it is used as an internal meta command to exit or quit the command line interface.



rm
It is used to remove a file in a directory.

